

Read mapping bonus exercise: IGV

Objective

You will use the web version of the Integrative Genomics Browser (IGV) to look at reads mapped to the human reference genome.

Step 1: Go to the IGV web app and load in a dataset

Go to: <https://igv.org/app/>

There are different versions of the human reference genome. For this exercise, we need the **GRCh37/hg19** version. Select this under “Genome”.

Then load in one of the example datasets, as follows: select “Tracks” -> “Platinum Genomes” -> “NA12877”. This loads an example dataset from the human sample NA12877. What’s being loaded is a BAM file with read alignments, and IGV allows those read alignments to be viewed visually.

You can now use the IGV browser to zoom in on different parts of chromosomes – once you zoom in close enough, you will see the reads being displayed as grey arrows along the genome. Hovering over a read with your mouse cursor will display various information about that read.

Step 2: Study a single-nucleotide polymorphism (SNP)

A single-nucleotide polymorphism (SNP) is a difference in a single DNA letter. A well-known SNP at human chromosome 2 is associated with lactase persistence: carriers of an A allele are able to digest the lactose sugar even into adulthood.

You can load in a track of known SNPs from the human population by selecting “Tracks” -> “Variants” -> “1KG Phase 3 Snps”.

Look at this position in IGV: chr2 : 136608646 . What is the genotype of the NA12877 sample at this position? Are they expected to be able to digest lactose as adults?

Step 2: Study a deletion

Put this locus into the IGV search bar: chr8 : 46,987,693–46,996,626

What’s going on here? Reads coloured **red** have a larger-than-expected distance between the two reads in a pair (see [IGV documentation](#) -> “Insert Size”).

Step 4: Study the amylase locus

First try to gain a sense for the average depth of coverage across the genome. Look at a few random locations in the genome. What is the coverage like? At the top there is a grey histogram displaying the coverage along the genome. You can also hover or click this coverage track to see the exact number at a given position.

Next, type in the gene name “AMY1A” into the search bar. What is the coverage like at this gene?

Look at the reads that are in white – they are coloured like this on the basis of their mapping quality. What is the mapping quality of the white reads? What does this mean?

Overall, what do you think is going on at this locus? If you’re interested, do some reading on amylase copy number variation.